

TITLE: GXP and Non-GXP Study Report Review and Approval Form

DOCUMENT NO. GE003FM01	VERSION 02	ISSUE DATE 17 Oct 2014	EFFECTIVE DATE 20 Oct 2014	PAGE 2 of 2
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☐ Vendor Study Protocol ☒ Vendor Study Report ☐ Lpath Study Report				
Lpath Study No.: LT3114-PHA-007-R		Version No.: 01	☒ Non GXP ☐ GCP	☐ GLP ☐ GMP
Product Name: LT3114		Test Article: LT3114; LT1023		
Study Title: Evaluation of 1023 and 3114 using the tail-flick test in the mouse (s.c. administration)				
Study Vendor/Site: Porsolt, France	☐ N/A	Study Director: Rosalia Matteo	☐ N/A	Study Contributor(s): Ann Gregus
Vendor Study No./Version: 14.0062	☐ N/A	Study Dates: July 2014	Report Date: September 2014	

Lab Notebook Reference: ☒ N/A

Executive summary of study results: ☐ See Attached ☐ N/A

The activity of LT 3114 was examined in a mouse model of nociception as measured by the latency of tail-flicking caused by thermal insult. LT 3114 (10 mg/kg and 50 mg/kg) was compared with a non-specific human IgG1 isotype control antibody (LT 1023; 50mg/kg) and was dosed subcutaneously once, 24 hours before the first test. Tail flick responses were measured at 24 h and 3 days after the single doses. Morphine (8mg/kg; s.c.) was administered as a positive control 30 minutes before testing on each of the test days (Day 1 and 3) in a separate group of animals. Morphine, as expected, was very effective at increasing the withdrawal latency compared with neutral vehicle controls (+89% and +71% increase on Days 1 and 3 respectively). However, neither LT3114 or LT 1023 demonstrated a significant analgesic effect at any of the doses administered.

LPATH APPROVAL

Function	Name (Print)	Signature	Date
Study Director/Monitor	Rosalia G Matteo	[Signature]	11/12/2014
Contributor	Ann M Gregus	[Signature]	11/12/2014
Research	Gary Woodman	[Signature]	11/12/2014
Development	Dario Pacciarini	[Signature]	11/18/14
Quality Assurance	ELVIRA MARQUEZ	[Signature]	26 NOV 2014

STUDY N° 14.0062
STUDY REPORT
EVALUATION OF 1023 AND 3114
USING THE TAIL-FLICK TEST IN THE MOUSE
(S.C. ADMINISTRATION)

Sponsor:

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<u>Study Plan Approval Date</u>	:	24 June 2014
<u>Experimental Start Date</u>	:	21 July 2014
<u>Experimental Termination Date</u>	:	24 July 2014
<u>First Draft Report Date</u>	:	18 August 2014
<u>Second Draft Report Date</u>	:	20 August 2014
<u>Final Report Date</u>	:	12 September 2014

TABULATED SUMMARY REPORT N° 14.0062	
<u>SPONSOR</u> : LPATH INC.	<u>SPONSOR STUDY REFERENCE</u> : Not communicated
<u>TEST SUBSTANCES</u> : 1023 and 3114	<u>TITLE</u> : Evaluation of 1023 and 3114 using the Tail-Flick Test in the mouse (s.c. administration)
<u>GENERIC NAME</u> : Not communicated	
<u>INDICATION</u> : Not communicated	<u>STUDY PERIOD</u> : July 2014 <u>FIRST DRAFT REPORT DATE</u> : 18 August 2014 <u>SECOND DRAFT REPORT DATE</u> : 20 August 2014 <u>FINAL REPORT DATE</u> : 12 September 2014
<u>AIM OF STUDY</u> : To evaluate the potential analgesic effects of 1023 and 3114 after s.c. administration using the Tail-Flick Test in the mouse.	
<u>ANIMALS</u>	: Male NMRI mice (Janvier Labs), 24-30 g (Day 0) body weight range.
<u>CONDITIONING OF ANIMALS</u>	: Stabilized for at least 5 days after delivery in macrolon cages (355 x 235 x 190 mm; 10 animals per cage) on wood litter with free access to food and water.
<u>TEST SUBSTANCES AND TREATMENT SCHEDULE</u>	: 1023 (Lpath Inc.), batch n° LP121424, colorless liquid, diluted in formulation buffer. 50 mg/kg, administered s.c. on Day 0 (24 hours before the first test) (10 ml/kg). Doses expressed in mg/kg of supplied substance. 3114 (Lpath Inc.), batch n° 103948, colorless liquid, diluted in formulation buffer. 10 and 50 mg/kg, administered s.c. on Day 0 (24 hours before the first test) (10ml/kg). Doses expressed in mg/kg of supplied substance.
<u>CONTROL SUBSTANCES</u>	: Formulation buffer (supplied by the sponsor) and phosphate buffered saline (neutral vehicle).
<u>REFERENCE SUBSTANCE</u>	: Morphine hydrochloride (Coopération Pharmaceutique Française, reference n° 1515300), batch n° D3065/2, dissolved in physiological saline. It was administered at 8 mg/kg s.c. 30 minutes before each test (10 ml/kg). Doses expressed in mg/kg of salt.
<u>METHOD OF EVALUATION</u>	: Effects in the test in comparison with vehicle controls.
<u>STATISTICAL ANALYSIS</u>	: One-way Anova followed by Dunnett's t test and Unpaired Student's t tests (main analysis) and Two-way Anova (complementary analysis).
<u>RESULTS:</u> <u>Baseline</u> Before dosing, the withdrawal latency was similar in all the groups. <u>Tests on Day 1 and Day 3</u> - Vehicle, administered s.c. one day before the first test, tended to decrease the withdrawal latency on Day 1 and Day 3, as compared with neutral vehicle controls (-20% and 34%, respectively) although this effect failed to reach statistical significance. - 1023 (50 mg/kg), administered s.c. one day before the first test, did not significantly affect the withdrawal latency on Day 1 or Day 3, as compared with vehicle controls. A slight tendency to increase the withdrawal latency was observed on Day 3 (+54%). - 3114 (10 and 50 mg/kg), administered s.c. one day before the first test, did not significantly affect the withdrawal latency on Day 1 or Day 3, as compared with vehicle controls. A dose dependent but non-significant tendency to increase the withdrawal latency was observed on Day 3 (+41% and +68 %, respectively). - Morphine (8 mg/kg), administered s.c. 30 minutes before the test on Day 1 and on Day 3, markedly increased the withdrawal latency, as compared with neutral vehicle controls (+89% and +71%, respectively, p < 0.001 for both).	
<u>CONCLUSION:</u> These results suggest the absence of significant analgesic activity for 1023 (50 mg/kg) and 3114 (10 and 50 mg/kg) after s.c. administration in the Tail-flick test in the mouse. A tendency to display analgesic effects was observed for the 2 substances as compared to vehicle controls on Day 3 but this interpretation is complicated by the withdrawal latencies of the vehicle control which are non-significantly lower than the corresponding latencies of neutral vehicle control.	
<u>STUDY CONDUCTED BY THE APPLICANT:</u> YES ☐ NO ☒	
<u>IF "NO", INDICATE THE NAME AND ADDRESS OF THE INSTITUTE THAT CONDUCTED THE STUDY:</u> Porsoft Research Laboratory, Z.A. de Glatigné, 53940 Le Genest-Saint-Isle, France	
<u>STUDY IN COMPLIANCE WITH GLP:</u> YES ☐ NO ☐ NOT REQUIRED ☒	

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1. REPORT APPROVAL

Camille Le CUDENNEC, Ph.D.
Study Director



Date: 12th September 2014

2. EXPERIMENTAL PROCEDURE

Tail-Flick Test in the mouse (reference PI 2M)

The method, which detects analgesic activity, follows that described by D'Amour and Smith (J. Pharmacol. Exp. Ther. 72, 74-79, 1941).

The mouse's tail was heated by means of a thermal light source (20 volts). The latency before the animal withdraws its tail was measured (maximum: 15 seconds). A baseline evaluation was performed in all mice before any dosing on Day 0.

1023 was evaluated at 50 mg/kg and 3114 was evaluated at 10 and 50 mg/kg. The test substances were administered s.c. on Day 0, 24 hours before the first test (Day 1), and compared with the vehicle control group (formulation buffer, group 2). A PBS control group was also added (group 1), this group was treated in similar conditions than group 2.

The animals were evaluated again 3 days post dosing (Day 3).

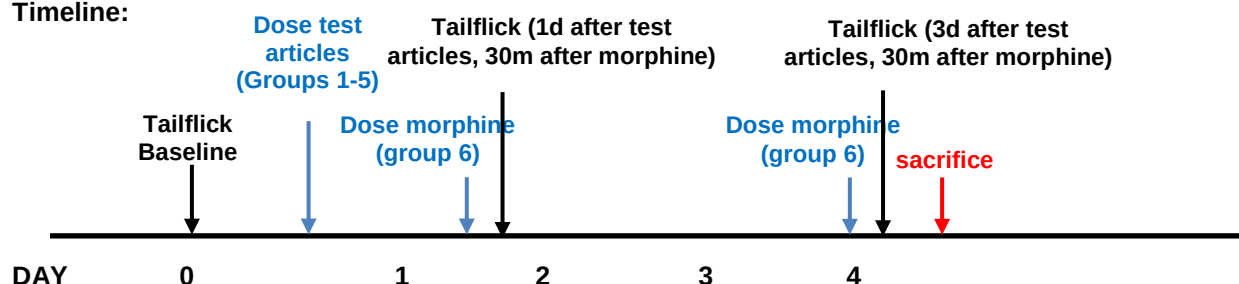
Morphine (8 mg/kg s.c.), administered 30 minutes before the test (Day 1), was used as reference substance. The same animals were evaluated again on Day 3, 30 minutes after a second administration of morphine. (Results were guaranteed in the first testing phase)

10 mice were studied per group. The test was performed partially blind.

The experiment therefore included 6 groups.

Group	Number of animals	Treatment	Dose-level (mg/kg)	Concentration (mg/ml)	Administration Volume (ml/kg)
1	10 males	PBS	0	0	10
2	10 males	Formulation buffer (vehicle)	0	0	10
3	10 males	1023	50	5	10
4	10 males	3114	10	1	10
5	10 males	3114	50	5	10
6	10 males	Morphine	8	0.8	10

Timeline:



Data with the test substance were analyzed by comparing treated groups with vehicle control using ANOVA followed by post-hoc Dunnett's tests. Data with the reference substance were analysed using unpaired Student's t tests.

A complementary analysis was performed using two-way ANOVA using treatment and time as factors (See Section 8, complementary statistical analysis).

3. RESULTS

Tail-Flick Test in the mouse

Baseline

Before dosing, the withdrawal latency was similar in all the groups.

Tests on Day 1 and Day 3

- Vehicle, administered s.c. one day before the first test, tended to decrease the withdrawal latency on Day 1 and Day 3, as compared with neutral vehicle controls (-20% and 34%, respectively) although this effect failed to reach statistical significance.
- 1023 (50 mg/kg), administered s.c. one day before the first test, did not significantly affect the withdrawal latency on Day 1 or Day 3, as compared with vehicle controls. A slight tendency to increase the withdrawal latency was observed on Day 3 (+54%).
- 3114 (10 and 50 mg/kg), administered s.c. one day before the first test, did not significantly affect the withdrawal latency on Day 1 or Day 3, as compared with vehicle controls. A dose dependent but non-significant tendency to increase the withdrawal latency was observed on Day 3 (+41% and +68 %, respectively).
- Morphine (8 mg/kg), administered s.c. 30 minutes before the test on Day 1 and on Day 3, markedly increased the withdrawal latency, as compared with neutral vehicle controls (+89% and +71%, respectively, $p < 0.001$ for both).

(See Section [Tables](#), [Figures](#) and [Individual Data](#))

4. CONCLUSION

These results suggest the absence of significant analgesic activity for 1023 (50 mg/kg) and 3114 (10 and 50 mg/kg) after s.c. administration in the Tail-flick test in the mouse. A tendency to display analgesic effects was observed for the 2 substances as compared to vehicle controls on Day 3 but this interpretation is complicated by the withdrawal latencies of the vehicle control which are non-significantly lower than the corresponding latencies of neutral vehicle control.

5. TABLES

TABLES**Tail-Flick Test in the mouse****Table 1: Body weight on Day 0 (g)**

Treatment	Body weight (g) (mean $\pm$ s.e.m.)
Neutral vehicle (s.c.)	27.3 $\pm$ 0.6
Vehicle (s.c.)	26.4 $\pm$ 0.6
1023 (50 mg/kg s.c.)	26.4 $\pm$ 0.4
3114 (10 mg/kg s.c.)	26.4 $\pm$ 0.4
3114 (50 mg/kg s.c.)	26.3 $\pm$ 0.5
Morphine (8 mg/kg s.c.)	26.0 $\pm$ 0.5

Tail-Flick Test in the mouse**Table 2: Effects of test substances on tail-Flick latency (s) - Baseline**

Treatment on Day 0	Tail-Flick latency (s) - Baseline		
	(mean $\pm$ s.e.m.)	Compared with Neutral vehicle (s.c.)	Compared with Vehicle (s.c.)
		(1) %	(1) %
Neutral vehicle (s.c.)	3.8 $\pm$ 0.6	NS +13%	
Vehicle (s.c.)	4.3 $\pm$ 0.6		
1023 (50 mg/kg s.c.)	4.8 $\pm$ 0.4	NS	+12%
3114 (10 mg/kg s.c.)	3.9 $\pm$ 0.5		-9%
3114 (50 mg/kg s.c.)	4.4 $\pm$ 0.5		+2%
Morphine (8 mg/kg s.c.) #	4.0 $\pm$ 0.5	NS +5%	

Table 3: Effects of test substances on tail-Flick latency (s) - Day 1

Treatment on Day 0	Tail-Flick latency (s) - Day 1		
	(mean $\pm$ s.e.m.)	Compared with Neutral vehicle (s.c.)	Compared with Vehicle (s.c.)
		(1) %	(1) %
Neutral vehicle (s.c.)	7.6 $\pm$ 1.2	NS -20%	
Vehicle (s.c.)	6.1 $\pm$ 1.3		
1023 (50 mg/kg s.c.)	7.2 $\pm$ 1.2	NS	+18%
3114 (10 mg/kg s.c.)	7.0 $\pm$ 1.5		+15%
3114 (50 mg/kg s.c.)	7.6 $\pm$ 1.2		+25%
Morphine (8 mg/kg s.c.) #	14.4 $\pm$ 0.4	*** +89%	

Table 4: Effects of test substances on tail-Flick latency (s) - Day 3

Treatment on Day 0	Tail-Flick latency (s) - Day 3		
	(mean $\pm$ s.e.m.)	Compared with Neutral vehicle (s.c.)	Compared with Vehicle (s.c.)
		(1) %	(1) %
Neutral vehicle (s.c.)	8.5 $\pm$ 1.5	NS -34%	
Vehicle (s.c.)	5.6 $\pm$ 1.2		
1023 (50 mg/kg s.c.)	8.6 $\pm$ 1.6	NS	+54%
3114 (10 mg/kg s.c.)	7.9 $\pm$ 1.3		+41%
3114 (50 mg/kg s.c.)	9.4 $\pm$ 1.2		+68%
Morphine (8 mg/kg s.c.) #	14.5 $\pm$ 0.3	*** +71%	

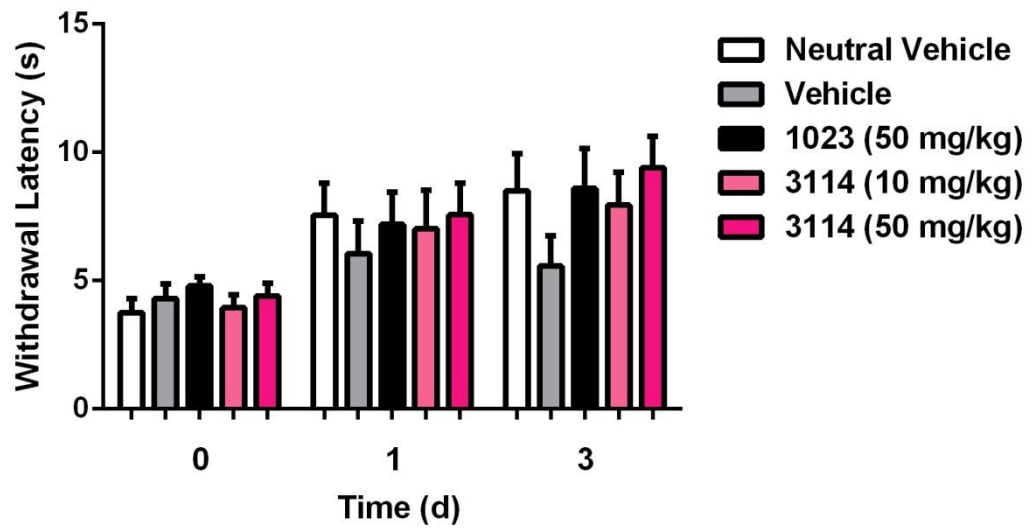
Inter-group comparison: NS = Not Significant; * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.

(1): For test substances treated groups: One-way ANOVA with group as factor.
For vehicle control and reference treated groups: Unpaired Student's t test.

#: -30 min on Days 1 and 3.

6. FIGURES (produced by the Sponsor)

Nociception



7. INDIVIDUAL DATA

INDIVIDUAL DATA**Tail-Flick Test in the mouse****Test substance..... : 1023**

Supplier..... : Lpath Inc
Batch number..... : LP121424
Appearance..... : colorless liquid
Preparation..... : diluted in formulation buffer (supplied by the sponsor)
Route of administration..... : s.c.
Pretreatment time..... : on Day 0 (24 h before the first test)
Dose expression..... : mg/kg of supplied substance
Volume of administration..... : 10 ml/kg body weight

Test substance..... : 3114

Supplier..... : Lpath Inc
Batch number..... : 103948
Appearance..... : colorless liquid
Preparation..... : diluted in formulation buffer (supplied by the sponsor)
Route of administration..... : s.c.
Pretreatment time..... : on Day 0 (24 h before the first test)
Dose expression..... : mg/kg of supplied substance
Volume of administration..... : 10 ml/kg body weight

Control substance..... : formulation buffer (supplied by the sponsor)

Route of administration..... : s.c.
Pretreatment time..... : on Day 0 (24 h before the first test)
Volume of administration..... : 10 ml/kg body weight

Reference substance..... : MORPHINE hydrochloride

Supplier..... : Coopération Pharmaceutique Française
Reference number..... : 1515300
Batch number..... : D3065/2
Appearance..... : white powder
Preparation..... : dissolved in physiological saline
Route of administration..... : s.c.
Pretreatment time..... : 30 min before each test on Day 1 and Day 3
Dose expression..... : mg/kg of salt
Volume of administration..... : 10 ml/kg body weight

Control substance..... : phosphate buffered saline (neutral vehicle)

Supplier..... : Sigma
Reference number..... : P-5368
Batch number..... : SLBH8322V
Route of administration..... : s.c.
Pretreatment time..... : on Day 0 (24 h before the first test)
Volume of administration..... : 10 ml/kg body weight

Animals..... : male NMRI mice (Janvier Labs)

Body weight range..... : 24-30 g (Day 0)
Delivery date..... : 15 July 2014
Number per group..... : 10
Experimental start date..... : 21 July 2014 (Day 0)
Experimental termination date..... : 24 July 2014 (Day 3)

Technician reference..... : CAL7-182

Tail-Flick Test in the mouse

Treatment on Day 0	Mouse number	Weight (g) (Day 0)	Tail-Flick latency (s)		
			Baseline	Day 1	Day 3
Neutral vehicle (s.c.)	D1	30	2.8	5.4	3.2
	D2	24	2.3	4.1	3.6
	D3	25	2.9	15.0	9.5
	D4	27	3.8	10.4	15.0
	D5	27	6.7	9.6	6.0
	D6	28	6.6	5.8	11.3
	D7	27	3.3	4.3	11.8
	D8	28	1.3	12.1	15.0
	D9	29	3.4	4.9	3.8
	D10	28	4.5	3.9	5.8
Mean ± s.e.m.		27.3 0.6	3.8 0.6	7.6 1.2	8.5 1.5

Treatment on Day 0	Mouse number	Weight (g) (Day 0)	Tail-Flick latency (s)		
			Baseline	Day 1	Day 3
Vehicle (s.c.)	E1	28	4.6	4.0	3.2
	E2	25	5.4	9.1	15.0
	E3	26	7.1	3.9	4.2
	E4	30	2.8	2.9	4.1
	E5	26	4.8	2.7	3.5
	E6	24	3.8	4.1	3.8
	E7	25	1.6	2.2	4.0
	E8	26	2.6	8.4	3.4
	E9	28	3.7	15.0	9.1
	E10	26	6.7	8.2	5.4
Mean ± s.e.m.		26.4 0.6	4.3 0.6	6.1 1.3	5.6 1.2
t value (a)			0.700	0.840	1.560
p value (a)			0.4932	0.4121	0.1362
			NS	NS	NS
Percent change (a)			+13%	-20%	-34%

Unpaired Student's t test: NS = Not Significant; * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$.

(a): compared with Neutral vehicle (s.c.).

Tail-Flick Test in the mouse

Treatment on Day 0	Mouse number	Weight (g) (Day 0)	Tail-Flick latency (s)		
			Baseline	Day 1	Day 3
1023 (50 mg/kg s.c.)	A1	27	3.8	5.8	5.7
	A2	28	6.3	2.4	15.0
	A3	25	5.9	4.2	5.3
	A4	27	3.9	13.4	15.0
	A5	26	3.2	14.3	15.0
	A6	25	4.2	6.1	9.3
	A7	27	4.5	6.8	9.7
	A8	28	5.7	7.8	4.9
	A9	25	4.1	7.9	3.0
	A10	26	6.3	3.3	3.0
Mean ± s.e.m.		26.4 0.4	4.8 0.4	7.2 1.2	8.6 1.6
Percent change (b)			+12%	+18%	+54%

Treatment on Day 0	Mouse number	Weight (g) (Day 0)	Tail-Flick latency (s)		
			Baseline	Day 1	Day 3
3114 (10 mg/kg s.c.)	C1	25	3.3	3.4	3.9
	C2	27	3.2	2.1	4.7
	C3	26	5.5	15.0	7.5
	C4	27	3.6	3.9	15.0
	C5	28	2.1	8.7	13.1
	C6	25	5.1	4.2	7.9
	C7	25	3.4	4.6	7.1
	C8	28	2.3	12.9	3.5
	C9	26	3.6	3.1	11.7
	C10	27	7.3	12.3	5.0
Mean ± s.e.m.		26.4 0.4	3.9 0.5	7.0 1.5	7.9 1.3
Percent change (b)			-9%	+15%	+41%

(b): compared with Vehicle (s.c.).

Tail-Flick Test in the mouse

Treatment on Day 0	Mouse number	Weight (g) (Day 0)	Tail-Flick latency (s)		
			Baseline	Day 1	Day 3
3114 (50 mg/kg s.c.)	B1	25	6.2	4.7	7.3
	B2	29	4.9	7.5	9.6
	B3	26	4.4	15.0	13.7
	B4	28	4.5	3.4	12.3
	B5	27	2.4	5.0	5.7
	B6	25	3.2	5.7	5.2
	B7	24	6.7	8.2	15.0
	B8	28	5.6	5.0	13.2
	B9	25	1.9	7.5	7.5
	B10	26	4.3	13.7	4.6
Mean ± s.e.m.		26.3 0.5	4.4 0.5	7.6 1.2	9.4 1.2
Percent change (b)			+2%	+25%	+68%

Treatment -30 min on Days 1 and 3	Mouse number	Weight (g) (Day 0)	Tail-Flick latency (s)		
			Baseline	Day 1	Day 3
Morphine (8 mg/kg s.c.)	F1	27	4.6	15.0	15.0
	F2	27	2.1	15.0	12.4
	F3	27	1.9	15.0	15.0
	F4	26	4.1	15.0	15.0
	F5	27	3.2	11.0	15.0
	F6	24	4.4	15.0	12.8
	F7	29	5.0	12.8	15.0
	F8	25	2.3	15.0	15.0
	F9	24	6.2	15.0	15.0
	F10	24	6.3	15.0	15.0
Mean ± s.e.m.		26.0 0.5	4.0 0.5	14.4 0.4	14.5 0.3
t value (a) p value (a)			0.333 0.7428 NS	5.187 <0.0001 ***	4.026 0.0008 ***
Percent change (a)			+5%	+89%	+71%

Unpaired Student's t test: NS = Not Significant; * = p < 0.05; ** = p < 0.01; *** = p < 0.001.

(a): compared with Neutral vehicle (s.c.).

(b): compared with Vehicle (s.c.).

8. COMPLEMENTARY STATISTICAL ANALYSIS (produced by the Sponsor)

Within each row, compare columns (simple effects within rows)

Number of families 3

Number of comparisons per family 4

Alpha 0.05

Bonferroni's multiple comparisons test Mean Diff. 95% CI of diff. Significant? Summary

0

Neutral Vehicle vs. Vehicle	-0.5500	-4.423 to 3.323	No	ns
Neutral Vehicle vs. 1023 (50 mg/kg)	-1.030	-4.903 to 2.843	No	ns
Neutral Vehicle vs. 3114 (10 mg/kg)	-0.1800	-4.053 to 3.693	No	ns
Neutral Vehicle vs. 3114 (50 mg/kg)	-0.6500	-4.523 to 3.223	No	ns

1

Neutral Vehicle vs. Vehicle	1.500	-2.373 to 5.373	No	ns
Neutral Vehicle vs. 1023 (50 mg/kg)	0.3500	-3.523 to 4.223	No	ns
Neutral Vehicle vs. 3114 (10 mg/kg)	0.5300	-3.343 to 4.403	No	ns
Neutral Vehicle vs. 3114 (50 mg/kg)	-0.02000	-3.893 to 3.853	No	ns

3

Neutral Vehicle vs. Vehicle	2.930	-0.9428 to 6.803	No	ns
Neutral Vehicle vs. 1023 (50 mg/kg)	-0.09000	-3.963 to 3.783	No	ns
Neutral Vehicle vs. 3114 (10 mg/kg)	0.5600	-3.313 to 4.433	No	ns
Neutral Vehicle vs. 3114 (50 mg/kg)	-0.9100	-4.783 to 2.963	No	ns

Test details Mean 1 Mean 2 Mean Diff. SE of diff. N1 N2 t DF

0

Neutral Vehicle vs. Vehicle	3.760	4.310	-0.5500	1.525	10	10	0.3608	108
Neutral Vehicle vs. 1023 (50 mg/kg)	3.760	4.790	-1.030	1.525	10	10	0.6756	108
Neutral Vehicle vs. 3114 (10 mg/kg)	3.760	3.940	-0.1800	1.525	10	10	0.1181	108
Neutral Vehicle vs. 3114 (50 mg/kg)	3.760	4.410	-0.6500	1.525	10	10	0.4264	108

1

Neutral Vehicle vs. Vehicle	7.550	6.050	1.500	1.525	10	10	0.9839	108
Neutral Vehicle vs. 1023 (50 mg/kg)	7.550	7.200	0.3500	1.525	10	10	0.2296	108
Neutral Vehicle vs. 3114 (10 mg/kg)	7.550	7.020	0.5300	1.525	10	10	0.3476	108
Neutral Vehicle vs. 3114 (50 mg/kg)	7.550	7.570	-0.02000	1.525	10	10	0.01312	108

3

Neutral Vehicle vs. Vehicle	8.500	5.570	2.930	1.525	10	10	1.922	108
Neutral Vehicle vs. 1023 (50 mg/kg)	8.500	8.590	-0.09000	1.525	10	10	0.05903	108
Neutral Vehicle vs. 3114 (10 mg/kg)	8.500	7.940	0.5600	1.525	10	10	0.3673	108
Neutral Vehicle vs. 3114 (50 mg/kg)	8.500	9.410	-0.9100	1.525	10	10	0.5969	108

Table Analyzed tailflick

Two-way RM ANOVA Matching: Across row
Alpha 0.05

Source of Variation	% of total variation	P value	P value summary	Significant?
Interaction	2.175 0.8448	ns	No	
Treatment	17.52 0.0002	***	Yes	
Time	2.706 0.2843	ns	No	
Subjects (matching)	20.30 0.1072	ns	No	

ANOVA table	SS	DF	MS	F (DFn, DFd)	P value
Interaction	47.64	8	5.955	F (8, 108) = 0.5124	P = 0.8448
Treatment	383.9	2	192.0	F (2, 27) = 11.65	P = 0.0002
Time	59.27	4	14.82	F (4, 108) = 1.275	P = 0.2843
Subjects (matching)	444.8	27	16.47	F (27, 108) = 1.417	P = 0.1072
Residual	1255	108	11.62		

Number of missing values 0

Within each row, compare columns (simple effects within rows)

Number of families 3

Number of comparisons per family 4

Alpha 0.05

Bonferroni's multiple comparisons test Mean Diff. 95% CI of diff. Significant? Summary

0

1023 (50 mg/kg) vs. Neutral Vehicle	1.030	-2.843 to 4.903	No	ns
1023 (50 mg/kg) vs. Vehicle	0.4800	-3.393 to 4.353	No	ns
1023 (50 mg/kg) vs. 3114 (10 mg/kg)	0.8500	-3.023 to 4.723	No	ns
1023 (50 mg/kg) vs. 3114 (50 mg/kg)	0.3800	-3.493 to 4.253	No	ns

1

1023 (50 mg/kg) vs. Neutral Vehicle	-0.3500	-4.223 to 3.523	No	ns
1023 (50 mg/kg) vs. Vehicle	1.150	-2.723 to 5.023	No	ns
1023 (50 mg/kg) vs. 3114 (10 mg/kg)	0.1800	-3.693 to 4.053	No	ns
1023 (50 mg/kg) vs. 3114 (50 mg/kg)	-0.3700	-4.243 to 3.503	No	ns

3

1023 (50 mg/kg) vs. Neutral Vehicle	0.09000	-3.783 to 3.963	No	ns
1023 (50 mg/kg) vs. Vehicle	3.020	-0.8528 to 6.893	No	ns
1023 (50 mg/kg) vs. 3114 (10 mg/kg)	0.6500	-3.223 to 4.523	No	ns
1023 (50 mg/kg) vs. 3114 (50 mg/kg)	-0.8200	-4.693 to 3.053	No	ns

Test details Mean 1 Mean 2 Mean Diff. SE of diff. N1 N2 t DF

0

1023 (50 mg/kg) vs. Neutral Vehicle	4.790	3.760	1.030	1.525	10	10	0.6756	108
1023 (50 mg/kg) vs. Vehicle	4.790	4.310	0.4800	1.525	10	10	0.3148	108
1023 (50 mg/kg) vs. 3114 (10 mg/kg)	4.790	3.940	0.8500	1.525	10	10	0.5575	108
1023 (50 mg/kg) vs. 3114 (50 mg/kg)	4.790	4.410	0.3800	1.525	10	10	0.2493	108

1

1023 (50 mg/kg) vs. Neutral Vehicle	7.200	7.550	-0.3500	1.525	10	10	0.2296	108
1023 (50 mg/kg) vs. Vehicle	7.200	6.050	1.150	1.525	10	10	0.7543	108
1023 (50 mg/kg) vs. 3114 (10 mg/kg)	7.200	7.020	0.1800	1.525	10	10	0.1181	108
1023 (50 mg/kg) vs. 3114 (50 mg/kg)	7.200	7.570	-0.3700	1.525	10	10	0.2427	108

3

1023 (50 mg/kg) vs. Neutral Vehicle	8.590	8.500	0.09000	1.525	10	10	0.05903	108
1023 (50 mg/kg) vs. Vehicle	8.590	5.570	3.020	1.525	10	10	1.981	108
1023 (50 mg/kg) vs. 3114 (10 mg/kg)	8.590	7.940	0.6500	1.525	10	10	0.4264	108
1023 (50 mg/kg) vs. 3114 (50 mg/kg)	8.590	9.410	-0.8200	1.525	10	10	0.5379	108